WAGER: Wealth-Anchored Gain Estimation of Reasoning

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Abstract

On the most label-imbalanced computer-vision benchmarks, a trivial predictor that looks up the most frequent label given cheap context already recovers most of the headline metric, so a reported gain from one method to another may reflect better fitting of annotation-frequency priors rather than genuine visual reasoning. Existing diagnostics are bespoke to a single benchmark and report point estimates with no uncertainty quantification. We introduce *WAGER* (Wealth-Anchored Gain Estimation of Reasoning), a distribution-free, *anytime-valid* framework that decomposes a benchmark gain into a prior-recoverable part and a genuine-reasoning part with a finite-sample validity certificate. The core construction is the *self-prior projection*: rather than comparing a model to an external frequency baseline, we collapse the model's *own* predictions onto the frequency-defining features and treat the result as a frozen competing forecaster. A single testing-by-betting capital process built on this projection simultaneously (i) certifies, via Ville's inequality, that the model uses image information beyond the prior (an e-value), and (ii) quantifies, via its growth rate, the model's usable reasoning information; the measurement is provably invariant to recalibration within a prior cell. Combining two such processes yields the *Reasoning Gain Ratio* (RGR) with an anytime-valid confidence interval and a decision rule (**RGR $<$ 0.5** flags a gain as non-substantive). On a simulation suite WAGER controls type-I error (0.017 at $\boldsymbol{\alpha}$**=0.05**), attains nominal interval coverage, and recovers an injected reasoning fraction with Pearson $\boldsymbol{r}$**=0.998**. On a real Visual Genome scene-graph-generation study over $\mathbf{7.6\times10^5}$ relations, the frequency baseline is correctly certified to perform no reasoning ($\boldsymbol{\hat{I}_{\text{reason}} \approx 0}$, e-value $\boldsymbol{\approx 1}$), and every accuracy gain over it is flagged non-substantive: the bulk of measured "information" is annotation-frequency prior-fitting, not reasoning. WAGER operates entirely on cached probability vectors, so the analysis is cheap and benchmark-agnostic.

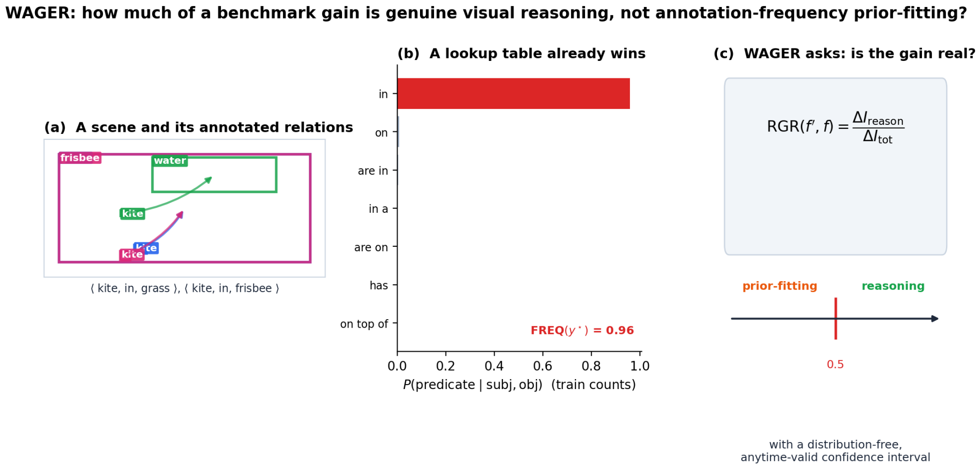

Fig. 1 The question addressed by WAGER. (a) A scene from Visual Genome and its annotated subject–predicate–object relations. (b) For a given ordered pair of subject and object classes, the training distribution $P(\text{predicate} \mid \text{subject}, \text{object})$ can be sharply concentrated on the annotated predicate. A lookup table can therefore make an accurate prediction without examining the image. (c) WAGER determines how much of an improvement between two methods arises from information beyond this frequency prior and reports the resulting Reasoning Gain Ratio with a distribution-free, anytime-valid confidence interval.

Keywords: Visual Reasoning, Anytime-Valid Inference, Dataset Bias, E-values, Scene Graph Generation, Visual Question Answering

1 Introduction

Progress on a visual benchmark is usually summarized by the difference between two headline scores. On strongly imbalanced datasets, however, this difference need not measure an improvement in visual reasoning. Annotation frequencies often provide a powerful prediction rule before the image is considered at all. In scene graph generation, for example, the FREQ baseline of Zellers et al. [1] predicts the most common predicate for each ordered subject–object class pair. On VG150 PredCls it reaches approximately 60.6 R@50, compared with 65.2 for MOTIFS, and thus recovers about 96% of the latter's recall through a lookup table. Figure 1 illustrates how this occurs for an individual Visual Genome annotation; in the full corpus, the five most common predicates account for 72% of all relations (Figure 3).

The same issue extends beyond scene graphs. In the original, unbalanced VQA data, answering “yes” without looking at the image achieves roughly 87% accuracy on questions beginning “Do you see a. . . ” [2]. When answer priors are deliberately changed in VQA-CP [3], UpDn falls from about 63% to 40%, and recent vision–language models remain measurably dependent on language priors [4, 5]. Frequency effects likewise dominate average precision for rare categories in LVIS [6]. These examples make the interpretation of a benchmark gain ambiguous: a model may have extracted more information from the image, or it may simply have fitted the annotation prior more effectively.

Existing responses expose important aspects of this problem but do not resolve that ambiguity for a particular pair of systems. Mean Recall@K [7, 8] changes the weight assigned to each category, yet it can reward suppression of head predictions and compromise category independence [9]. Changing-prior splits such as VQA-CP can themselves be exploited by reversing known label statistics; on some question types, a random-answer strategy can outperform the state of the art [10, 11]. Causal total-direct-effect inference [12] and more recent debiasing methods [13] intervene during training or inference rather than measuring the source of an observed gain. More generally, these approaches are tied to particular benchmarks and return point estimates without uncertainty quantification. They therefore cannot answer whether a specified improvement is primarily attributable to reasoning, nor with what confidence.

We address this gap with WAGER (Wealth-Anchored Gain Estimation of Reasoning), a distribution-free framework for separating a benchmark gain into a prior-recoverable component and a residual reasoning component. Its central device is a *self-prior projection*: the predictions of a model are averaged over the features that carry the annotation-frequency prior. The resulting forecaster is the same model with instance-level visual information removed. Comparing a model with this frequency-collapsed version of itself avoids the unknown mismatch introduced by an external frequency baseline and makes the comparison invariant to calibration shared within each prior cell.

WAGER turns that predictive comparison into a testing-by-betting process. The capital accumulated by the process has two complementary interpretations. At any sample size it is an e-value that certifies whether the model predicts beyond its self-prior projection; asymptotically, its growth rate measures the model's usable reasoning information. Applying the construction to two models yields the Reasoning Gain Ratio (RGR), the fraction of their total information gain attributable to the reasoning channel. An anytime-valid confidence interval supports an operational decision rule: when the upper confidence limit is below 0.5, most of the gain is prior-recoverable and the improvement is flagged as non-substantive.

This work makes four contributions. First, we introduce the self-prior projection (Definition 1), which provides an exact predictive null without access to the true prior and yields calibration invariance (Corollary 1). Second, we show that one capital process both certifies and quantifies reasoning beyond that projection (Theorems 1–2). Third, we define the RGR and derive its distribution-free, anytime-valid confidence interval (Definition 4; Theorem 3). Finally, we validate these properties in simulation and conduct a Visual Genome study containing 7.6×10^5 relations. The simulations show type-I control, nominal interval coverage, and accurate recovery of injected reasoning. In the real-data study, WAGER correctly identifies the frequency baseline as non-reasoning and flags every accuracy gain over it as non-substantive. Because the analysis consumes cached probability vectors, it can be run in minutes on a CPU after logits have been generated.

2 Related Work

WAGER sits at the intersection of four largely separate literatures: the study of dataset bias and shortcut learning, the benchmark-specific critiques and debiasing methods developed for scene graph generation and visual question answering, the predictive notion of usable information, and the game-theoretic statistics of e-values and anytime-valid inference. We review each in turn, emphasising throughout the distinction between work that *mitigates* a bias or *diagnoses* a benchmark and our own goal, which is to *attribute a specific reported gain* to reasoning or prior-fitting with a finite-sample certificate.

Dataset bias and shortcut learning.

The tendency of models to exploit superficial statistics rather than the intended signal is long documented [14, 15] and was framed as a general failure mode by Geirhos et al. [16], who argued that shortcut features are the rule rather than the exception when a cheap cue is predictive on the training distribution. The standard responses—reweighting, resampling, group distributionally-robust training, or building out-of-distribution splits—change *how models are trained or which test set is used*. They are valuable for building more robust systems, but they do not, and are not intended to, deliver a certified measurement of how much of a particular reported gain between two existing systems is shortcut. WAGER targets exactly that measurement and leaves training untouched.

Debiasing and evaluation in scene graph generation.

Frequency priors dominate scene graph generation, motivating a large body of debiasing work: re-weighted and re-sampled losses, dynamic tree contexts [8], knowledge routing [7], causal total-direct-effect inference [12], and recent informative-SGG methods [13]. The very fact that this stream of methods remains active is evidence that the underlying confound is not resolved, only attenuated. On the evaluation side, mean Recall@K [7, 8] and its independent-mean refinement [9] rebalance metric weight so that rare predicates count more, but they remain point estimates with no uncertainty quantification and can be gamed by suppressing head predictions to inflate the mean. WAGER is complementary to both lines: it is an *evaluation primitive* that attributes a gain to a channel and reports an anytime-valid interval, rather than a training method that improves tail recall or a re-weighted scalar that can itself be optimised against. Its head/body/tail stratification (Section 4.4) speaks to the same tail-collapse concern these metrics target, but measures usable reasoning per tier instead of re-weighting recall.

Language priors in VQA and multimodal LLMs.

The “blind” solvability of VQA [2] prompted changing-prior splits [3], which were then shown to be exploitable by label inversion [10, 11]. The problem persists for modern vision-language models: benchmarks such as VLind-Bench [4] and analyses like LanP [5] document substantial reliance on language priors, and recent work continues to develop debiasing methods [17]. These efforts diagnose a benchmark or fix a model;

WAGER instead certifies and quantifies a *gain between two systems* with a validity guarantee, and instantiates uniformly across modalities through the choice of prior feature ϕ (Section 4).

Long-tail recognition and detection.

Closely related is long-tailed object detection, where rare-category average precision is dominated by frequency priors [6] and recent work targets the resulting regression and classification bias [18]. WAGER's head/body/tail stratification (Section 4.4) speaks directly to this literature, but measures *usable reasoning per tier* rather than re-weighting a metric.

Benchmark reliability and contamination.

A broader reliability crisis now surrounds static benchmarks: surveys and analyses document that leaderboard gains can reflect data contamination and memorization rather than capability, especially for multimodal models where both images and text may leak [19–21]. The principal response has been to develop *dynamic* or refreshed benchmarks. WAGER addresses a complementary question: it leaves the benchmark fixed and asks, with a certificate, whether a specific reported gain reflects information beyond a stated confound.

Usable information.

Our reasoning channel measures *usable* (not information-theoretic) information, in the predictive, family-relative sense of $\mathcal{V}$-information [22, 23], conditional probing [24, 25], and usable-information dataset diagnostics [26]. WAGER's novelty is to measure this quantity *beyond a self-induced prior baseline* and to attach an anytime-valid certificate to it.

Game-theoretic statistics and anytime-valid inference.

WAGER is built on testing-by-betting and e-values [27–30], the bounded-mean capital process of Waudby-Smith and Ramdas [31], and the forecaster-comparison e-process of Henzi and Ziegel [32]; recent work has further developed the foundations and methodology [33–35] and extending e-values to applied sequential monitoring [36]. The growth-rate–information identity goes back to Kelly [37] and the universal-portfolio view [38, 39]. To our knowledge WAGER is the first to cast benchmark-confound attribution as a game-theoretic e-process, yielding a single object that both certifies (Theorem 1) and quantifies (Theorem 2) reasoning beyond annotation-frequency priors. Unlike conditional-independence testing, which is provably hard in general [40], we run an honest predictive comparison of two fixed forecasters, which is exactly what betting certifies.

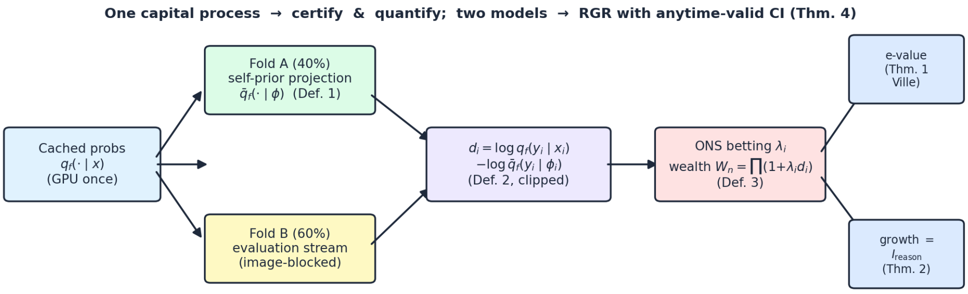

Fig. 2 The WAGER pipeline. Cached probabilities are split image-blocked into a projection fold A and an evaluation stream B. The self-prior projection $\bar{q}_f$ is frozen on A (Def. 1); on B the per-instance prior-excess score d_i (Def. 2) drives an Online-Newton-Step betting process whose wealth W_n (Def. 3) is read two ways: as an e-value (Thm. 1) and as a growth rate equal to usable reasoning information (Thm. 2). Two models combine into the RGR with an anytime-valid CI (Thm. 3).

3 Method

3.1 Setup and the self-prior projection

Let x be an instance (for scene graph generation, an ordered object pair and its image region), $y \in \{1, \ldots, K\}$ the ground-truth label, and $\phi(x)$ the *prior features*: the cheap, frequency-carrying sufficient statistic through which annotation frequency acts (for SGG, $\phi(x) = (\text{subject class}, \text{object class})$). A model f supplies a predictive distribution $q_f(y \mid x)$, the softmax of cached logits, and $u(y) = 1/K$ is the uniform reference. We never assume that q_f is calibrated, nor that the true conditionals $p(y \mid x)$ or $p(y \mid \phi)$ are known. The only structural assumption is that instances on an evaluation stream B are *exchangeable*, which we enforce by randomizing their order with image-blocked permutations (Section 4).

The object at the centre of the method collapses a model onto its own prior.

Definition 1 (Self-prior projection)**.** For a fixed model f, its self-prior projection is the model's own predictions averaged within each prior-feature cell,

$$\bar{q}_f(y \mid \phi) \;=\; \mathbb{E}[\, q_f(y \mid X) \mid \phi(X) = \phi\,]\,. \tag{1}$$

By construction $\bar{q}_f$ is the best summary of f that is measurable with respect to the prior features alone: a forecaster that has discarded every pixel-level cue and kept only what frequency context implies. The decisive choice is to compare f with this frequency-collapsed version of *itself* rather than with an external frequency table. An external table $\hat{p}(y \mid \phi)$ built from training counts would make the null "f reasons no better than the prior" hold only if the table equalled the true $p(y \mid \phi)$, which it never exactly does; that mismatch is a free parameter an adversary or an unlucky dataset could exploit to manufacture or hide reasoning. Freezing $\bar{q}_f$ from f's own predictions removes the parameter, because under the null the two forecasters share every nuisance—calibration, vocabulary bias, temperature—so any out-of-fold predictive advantage of q_f over $\bar{q}_f$ can come only from information beyond ϕ.

We estimate $\bar{q}_f$ on a fold A disjoint from the evaluation stream. A naive cell mean is unusable when most cells are sparse (Figure 3), so we use a hierarchical empirical-Bayes backoff with Krichevsky–Trofimov-style smoothing: writing n_ϕ for the cell count and $\pi(\phi)$ for a parent distribution,

$$\hat{\bar{q}}_f(\cdot \mid \phi) \;=\; \frac{\sum_{j \in A} q_f(\cdot \mid x_j)\mathbb{1}[\phi(x_j) = \phi] \;+\; m\,\pi(\phi)}{n_\phi + m}, \tag{2}$$

with the hierarchy cell $(\mathrm{subj}, \mathrm{obj}) \to$ coarse $(\mathrm{subj}) \to$ global mean. A second-order Jensen correction computed from the within-cell variance debiases the log-projection; it cancels in the total information I^{tot} and only re-allocates mis-credited reasoning back into the prior channel. A model whose predictions are a function of ϕ alone—FREQ, or a class-only network—uses the identical backoff, so its out-of-fold reasoning score collapses to ≈ 0, which is exactly what makes the FREQ anchor hold on real data (Section 4).

3.2 Certifying and quantifying reasoning by betting

On the evaluation stream we score each instance by how much the model beats its own projection, and then bet on those scores.

Definition 2 (Prior-excess log-score)**.** On the evaluation stream, define

$$d_i \;=\; \log q_f(y_i \mid x_i) - \log \hat{\bar{q}}_f(y_i \mid \phi(x_i)), \qquad \tilde{d}_i = \mathrm{clip}(d_i, -c, c), \tag{3}$$

with default clip $c = \log K$. A positive d_i means that at instance i, using the image beyond ϕ helped predict the true label. Clipping keeps the score bounded, which is what licenses the betting argument below; the level c trades robustness against sensitivity, and we fix it at $\log K$, the natural scale of a log-likelihood ratio over K classes.

Definition 3 (Wealth process)**.** We test $H_0 : \;\mu := \mathbb{E}[\tilde{d}_i] \le 0$ ("f predicts no better than its prior-projection") against $H_1 : \mu > 0$ via

$$W_0 = 1, \qquad W_n = \prod_{i=1}^{n}\bigl(1 + \lambda_i\,\tilde{d}_i\bigr), \qquad \lambda_i \in [0, \tfrac{1}{c}) \text{ predictable}, \tag{4}$$

with λ_i chosen online by the Online-Newton-Step (aGRAPA) rule of Waudby-Smith and Ramdas [31], a closed-form predictable update requiring no tuning.

Framing the question as a bet on a predictive mean, rather than as a test of the conditional independence $Y \perp X \mid \phi(X)$, is deliberate: consistent conditional-independence testing is impossible without further assumptions [40], whereas testing

$\mathbb{E}[\tilde{d}] \leq 0$ with a betting martingale is precisely the regime in which game-theoretic statistics delivers finite-sample validity [27, 29, 31].

Theorem 1 (Anytime-valid certificate)**.** *Under H_0, (W_n) is a non-negative supermartingale with $\mathbb{E}[W_n] \leq 1$; it is therefore an e-value, and by Ville's inequality [41], for any $\alpha \in (0,1)$, $\Pr[\exists n : W_n \geq 1/\alpha] \leq \alpha$. Rejecting H_0 the first time $W_n \geq 1/\alpha$ controls type-I error at level α with no fixed sample size and no distributional assumption beyond exchangeability.*

Proof sketch λ_i is $\mathcal{F}_{i-1}$-measurable and $1 + \lambda_i \tilde{d}_i \geq 0$ because $\lambda_i < 1/c$ and $\tilde{d}_i \geq -c$. Under H_0, $\mathbb{E}[1 + \lambda_i \tilde{d}_i \mid \mathcal{F}_{i-1}] = 1 + \lambda_i \mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}] \leq 1$, so (W_n) is a non-negative supermartingale; telescoping gives $\mathbb{E}[W_n] \leq 1$ and Ville's inequality yields the bound. The validity holds *without knowing the true prior* precisely because $\bar{q}_f$ is frozen on fold A and treated as a fixed competing forecaster. A full proof is given in Appendix A. □

Theorem 2 (Growth rate = usable reasoning information)**.** *With the log-optimal predictable bet $\lambda_i^\star = \arg\max_\lambda \mathbb{E}[\log(1 + \lambda \tilde{d}) \mid \mathcal{F}_{i-1}]$, the growth rate converges almost surely, and as $c \to \infty$,*

$$G(f) := \lim_{n\to\infty} \frac{1}{n} \log W_n = \mathbb{E}[\log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi(X))] =: \mathcal{I}_f^{\text{reason}} \geq 0. \quad (5)$$

If q_f equals the true posterior and $\bar{q}_f$ the true $p(\cdot \mid \phi)$, then $\mathcal{I}_f^{\text{reason}} = I(Y; X \mid \phi(X))$, the conditional mutual information.

Proof sketch By Kelly [37] and Cover and Thomas [39] the log-optimal growth rate equals the expected log-score differential, which the strong law drives to $\mathbb{E}[\tilde{d}]$; as $c \to \infty$, $\tilde{d}_i \to d_i$. The chain rule $I(Y; X) = I(Y; \phi(X)) + I(Y; X \mid \phi(X))$, valid because ϕ is a function of X, identifies the residual with conditional mutual information in the calibrated case (Appendix A). □

The same wealth process therefore *certifies* reasoning (through W_n, Theorem 1) and *quantifies* it (through $G(f)$, Theorem 2)—the central unification of the method. The measurement is moreover robust to miscalibration.

Corollary 1 (Calibration invariance)**.** *$\mathcal{I}_f^{\text{reason}}$ is invariant to any recalibration map applied uniformly within a ϕ-cell (e.g. per-cell temperature scaling), because the map cancels between q_f and $\bar{q}_f$. Hence miscalibration shared within a prior cell cannot inflate measured reasoning; by Gibbs' inequality an arbitrarily miscalibrated q_f can only* under-*estimate the true reasoning information, never manufacture it.*

3.3 The Reasoning Gain Ratio

We decompose each model's total information over the uniform reference into a prior-fitting channel and a reasoning channel,

$$\underbrace{\mathbb{E}[\log \bar{q}_f(Y \mid \phi) - \log u(Y)]}_{I_f^{\text{prior}}} + \underbrace{\mathbb{E}[\log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi)]}_{I_f^{\text{reason}}} = I_f^{\text{tot}}, \tag{6}$$

and attribute a reported improvement to one channel or the other.

Definition 4 (Reasoning Gain Ratio)**.** For a claimed improvement from f to f',

$$\mathrm{RGR}(f', f) = \frac{I_{f'}^{\text{reason}} - I_f^{\text{reason}}}{I_{f'}^{\text{tot}} - I_f^{\text{tot}}}, \tag{7}$$

the fraction of the information gain attributable to better reasoning rather than better prior-fitting. If the upper confidence limit of RGR is below 0.5, the majority of the gain is prior-recoverable and the claim is flagged *non-substantive*.

Theorem 3 (Anytime-valid CI for RGR)**.** *Each of* I^{reason} *and* I^{tot} *is the mean of a bounded score, so the betting confidence sequence of [31] gives each an anytime-valid* $(1-\alpha)$ *interval. Combining the numerator and denominator intervals by Fieller's method [42] under a Bonferroni split yields a* $(1-2\alpha)$ *anytime-valid interval for* $\mathrm{RGR}(f', f)$.

3.4 Algorithm and implementation

Algorithm 1 summarizes the driver. Two properties make it both cheap and exact. First, the GPU is used only once, to dump the cached probability vectors; all WAGER computation is deterministic CPU NumPy with an $O(NK)$ hot loop, so the full five-model study including R permutations and every interval completes in about three minutes on a single core. Second, because the per-instance scores are order-independent under a frozen train-fit projection, the R image-blocked permutations only reshuffle the betting order, which builds the e-value distribution and averages the anytime-valid intervals across orders. Blocking at the image level—all relations of an image move together in randomized image order—yields an exchangeable stream at the block level, which is all Theorem 1 requires.

A concrete instance fixes intuition. Consider a cell $\phi = (\text{man}, \text{horse})$ whose frozen projection assigns $\bar{q}_f(\text{riding} \mid \phi) = 0.30$. If, for an instance whose true predicate is "riding", the model sees the boxes overlap with the man above the horse and outputs $q_f(\text{riding} \mid x) = 0.62$, then $d = \log 0.62 - \log 0.30 = +0.73 > 0$ and the bettor profits, because the image moved the prediction toward the truth beyond what the class pair alone implies. If instead the model returns the cell-typical 0.30, then $d = 0$ and the bettor neither gains nor loses. A model that is merely *confident* but constant within the cell earns no capital at all—the mechanism behind both the FREQ anchor and calibration invariance (Corollary 1).

Algorithm 1 WAGER core routine

Require: cached q_f for each model; prior map ϕ; coarse key; level α; clip $c = \log K$; permutations R.

Ensure: per-model e-value and $\hat{I}^{\text{reason}}, \hat{I}^{\text{prior}}$ with CIs; per-pair RGR with verdict.

fit frozen $\bar{q}_f$ on fold A via hierarchical backoff (Def. 1)
for each model f and each i in stream B **do**
  $d_i \leftarrow \text{clip}(\log q_f(y_i \mid x_i) - \log \bar{q}_f(y_i \mid \phi_i), -c, c)$; $e_i \leftarrow \text{clip}(\log \bar{q}_f(y_i \mid \phi_i) - \log u(y_i), -c, c)$
end for
for $r = 1 \dots R$ (image-blocked permutations of B) **do**
  $\lambda_i \leftarrow \text{ONS}(d_{1:i-1})$; $W_n \leftarrow \prod(1 + \lambda_i d_i)$ ▷ e-value, growth (Thm. 1,2)
end for
$\hat{I}^{\text{reason}} \leftarrow \text{mean}(d)$, $\hat{I}^{\text{prior}} \leftarrow \text{mean}(e)$, with betting CIs
for each pair: RGR $\leftarrow \Delta\hat{I}^{\text{reason}} / \Delta\hat{I}^{\text{tot}}$ with Fieller CI (Thm. 3); verdict by the 0.5 rule

4 Experiments

Our evaluation proceeds from controlled validation to a real-data application. We first describe the corpus, the prior-feature instantiation, and the models (Section 4.1). We then test WAGER in simulations for which the true reasoning contribution is known (Section 4.2), apply it to scene graph generation on Visual Genome (Sections 4.3–4.4), and finally ask whether the conclusions depend on the principal design choices (Section 4.5). Unless otherwise noted, all intervals use $\alpha = 0.05$.

4.1 Datasets and experimental setup

Our study uses the Visual Genome scene-graph corpus [43] in the PredCls regime—ground-truth boxes and object labels are given and the predicate is to be predicted—with the standard VG150 vocabulary of 150 object classes and 50 predicates. Table 1 summarizes the corpus. Two properties make it an ideal stress test for a tool that separates reasoning from prior-fitting. First, the predicate distribution is heavily *long-tailed*: the five most frequent predicates account for 71.6% of all relation instances and the single predicate "on" for 32.4% (Figure 3), so a frequency lookup already secures most of any recall-style metric. Second, the prior-feature space is *sparse*: only 49.2% of the 150×150 possible ordered class pairs are observed in training, and 46.4% of the observed cells contain fewer than five instances. This sparsity is exactly why a naive cell mean fails and the hierarchical backoff of Section 3 is required to keep the self-prior projection honest. We hold out half of the test images, image-blocked, as the frozen projection fold (113,255 relations) and bet on the remaining 116,350.

WAGER is instantiated through the prior feature ϕ, the statistic through which annotation frequency acts. For Visual Genome PredCls ϕ is the ordered class pair, but the same machinery applies to VQA/GQA with ϕ the question type and to LVIS with ϕ the frequency tier (Table 2); this is the sense in which the method is benchmark-agnostic, and we return to those settings in Section 5.

Table 1 Visual Genome PredCls corpus statistics. The test images are split image-blocked into a projection fold (for $\bar{q}_f$) and an evaluation stream (for betting).

Quantity	Value
Relation instances (total)	762,592
training / test	532,987 / 229,605
test: projection / eval fold	113,255 / 116,350
Images (train / test)	65,228 / 27,955
Relations per image (mean / median, test)	8.21 / 6
Predicate classes K	50
Object classes	150
Prior-feature cells (observed / possible)	11,076 / 22,500
cells with < 5 training instances	46.4%
Top-5 predicate share / "on" share	71.6% / 32.4%
Head / body / tail predicates (#)	5 / 11 / 34
Head / body / tail instance share	71.6% / 16.6% / 11.8%

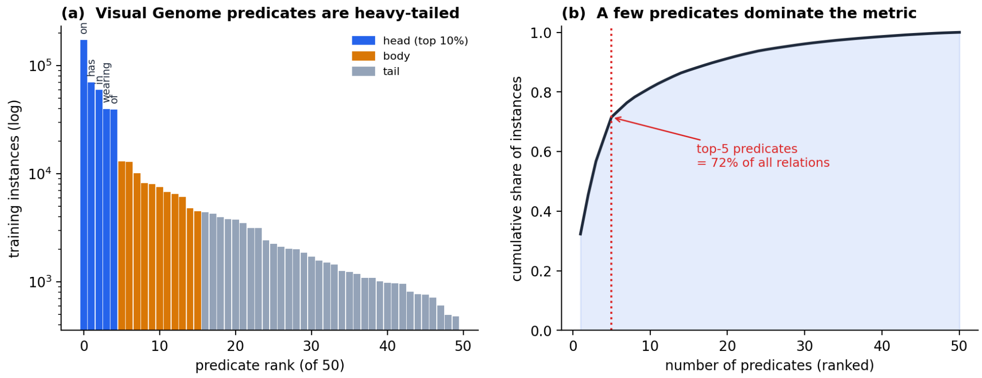

Fig. 3 **Predicate-frequency imbalance.** (a) Visual Genome's 50 predicates follow a heavy-tailed training distribution (log scale; head/body/tail). (b) The five most frequent predicates account for 72% of all relation instances, so a frequency lookup already secures most of the recall metric.

To exercise the method we construct five predicate predictors that expose progressively more information beyond the ordered class pair. FREQ is the training-count lookup $P(\text{predicate} \mid \text{subject}, \text{object})$ and is therefore constant within a ϕ-cell. FREQ+OVERLAP supplements it with a binary indicator of box overlap, while MLP-CLASS learns from class embeddings alone and so remains essentially ϕ-measurable. MLP-SPATIAL additionally observes translation- and scale-invariant box geometry—relative position, scale, aspect ratio, and intersection over union—and MLP-SPATIAL+ increases the capacity of that spatial model. The geometric measurements are the information beyond ϕ that the study uses to separate reasoning from prior-fitting. Each model is trained once, after which WAGER operates only on its cached predicate distributions.

Table 2 Prior feature ϕ per benchmark. The empirical study in this paper uses the first row.

Benchmark	Task	Prior feature $\phi(x)$
Visual Genome (VG150)	Scene graph generation	(subject class, object class)
GQA / VQA v2	Visual question answering	question type / template
LVIS	Long-tail detection	category-frequency tier

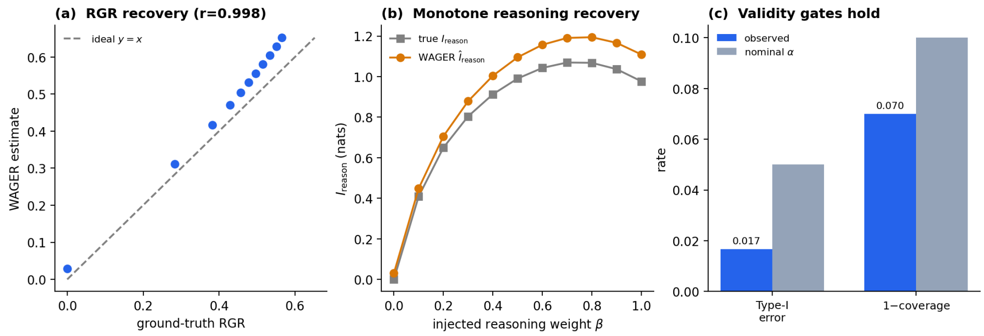

Fig. 4 **Phase-1 validation.** (a) Estimated vs. ground-truth RGR ($r = 0.998$). (b) Monotone recovery of injected reasoning information. (c) Type-I error and one-minus-coverage stay within their nominal levels.

4.2 Simulation validity and power

We begin with synthetic data because it permits direct tests of validity and power against known population quantities. A prior forecaster is mixed with an oracle as $q_f = (1 - \beta)\,\text{prior} + \beta\,\text{oracle}$, making β a controlled source of within-cell information. At the null, where $q_f = \bar{q}_f$, the wealth process exceeds the rejection threshold in 0.017 of 600 runs—below the nominal $\alpha = 0.05$—while $\hat{I}^{\text{reason}} \approx 0.004$, as predicted by Theorem 1. The anytime-valid intervals attain empirical coverage of 0.93 against a 0.90 target. As β increases, the estimated RGR changes monotonically and follows its ground-truth value with Pearson $r = 0.998$, and the correlation between estimated and true reasoning information is $r = 0.999$. Figure 4 presents the three results together. Thus, before any real-data claim is made, the simulations establish type-I control, interval coverage, and faithful recovery of the quantity WAGER is intended to measure.

4.3 Results on Visual Genome

Table 3 first confirms the anchor behaviour the framework requires. FREQ returns $\hat{I}^{\text{reason}} = -0.005$, with a confidence interval straddling or below zero, and an e-value of $0.49 \approx 1$: WAGER correctly certifies that a lookup table performs no visual reasoning, and FREQ+OVERLAP is likewise certified non-reasoning. The MLPs that genuinely consult box geometry instead accumulate decisive evidence, with e-values reaching the numerical ceiling $\sim 10^{304}$ and positive $\hat{I}^{\text{reason}}$ that grows with spatial capacity. Test

Table 3 WAGER readouts on Visual Genome PredCls (eval $N = 116{,}350$; $\alpha = 0.05$). I^{reason} shown with its anytime-valid CI. Test accuracy moves by under 1.3 points across all five models, yet the e-value and reasoning channel separate them sharply.

Model	e-value	$\hat{I}^{\text{reason}}$ (CI)	I^{prior}	I^{tot}	acc.
FREQ	0.49	$-0.005\,[-0.006, -0.001]$	2.524	2.518	0.6564
FREQ+OVERLAP	0.12	$-0.017\,[-0.018, -0.011]$	2.477	2.460	0.6545
MLP-CLASS	$4{\times}10^{145}$	$+0.028\,[+0.024, +0.030]$	2.688	2.716	0.6557
MLP-SPATIAL	$> 10^{300}$	$+0.063\,[+0.056, +0.066]$	2.699	2.762	0.6639
MLP-SPATIAL+	$> 10^{300}$	$+0.071\,[+0.063, +0.074]$	2.710	2.781	0.6674

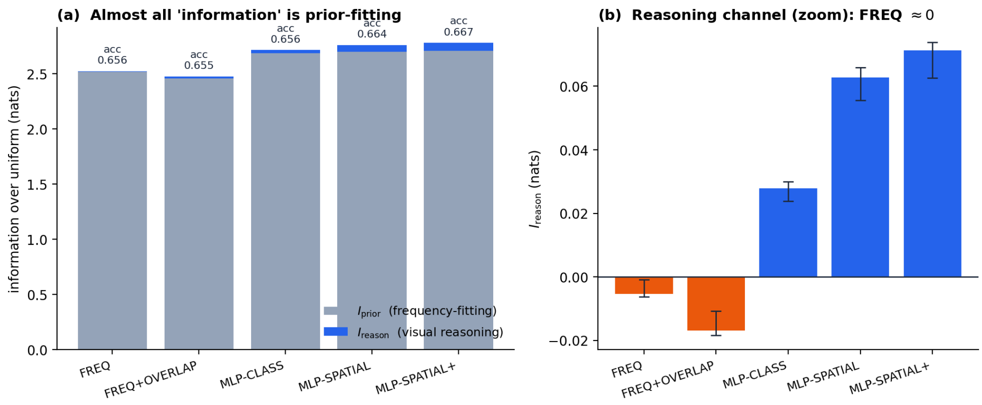

Fig. 5 **Information decomposition.** (a) Stacked I^{prior} (slate) and I^{reason} (blue) per model with test accuracy; the reasoning sliver is barely visible against prior-fitting. (b) The reasoning channel alone, with anytime-valid CIs: FREQ/FREQ+OVERLAP are ≤ 0, the spatial MLPs are clearly positive.

accuracy, by contrast, varies by less than 1.3 percentage points across all five systems, so it gives almost no indication of whether an improvement arises from reasoning or from prior-fitting—the distinction WAGER is built to make.

The information decomposition in Figure 5 explains the discrepancy between accuracy and certified reasoning. For every model the prior channel ($I^{\text{prior}} \approx 2.5$ nats) dwarfs the reasoning channel ($I^{\text{reason}} \leq 0.07$ nats): even the strongest model spends more than 97% of its information budget on frequency-fitting. The testing-by-betting view in Figure 6 makes the same point dynamically—the capital of FREQ and FREQ+OVERLAP never leaves the neighbourhood of its starting value, while the spatial MLPs compound capital linearly on a log scale, the signature of a model that genuinely out-predicts its own self-prior projection. Figure 7 resolves the mechanism inside a single ϕ-cell: the projection $\bar{q}_f$ is the frozen cell forecaster, and it is the spread of per-instance scores d_i around it that the betting process converts into capital.

These aggregate readouts have a concrete instance-level counterpart on the images themselves. Figure 8 overlays our subject/object boxes and relation arrows on the actual Visual Genome images we process, annotating each relation with its predicted

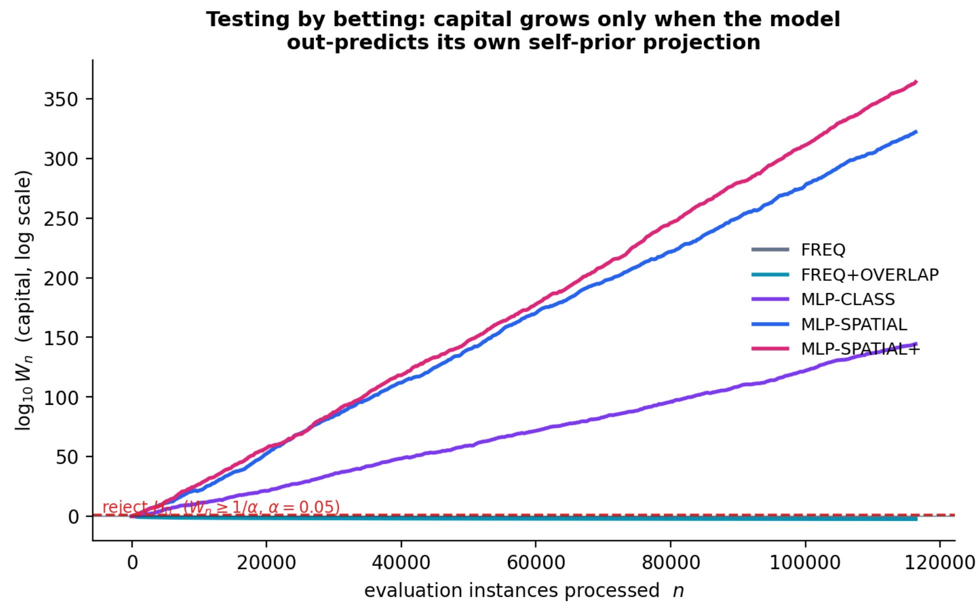

Fig. 6 Testing by betting. Log-capital $\log_{10} W_n$ over the evaluation stream. FREQ and FREQ+OVERLAP stay flat (no evidence of reasoning); the spatial MLPs compound capital and cross the rejection threshold $1/\alpha$ almost immediately.

predicate, FREQ's top guess, and the score d_i for MLP-SPATIAL. Where the predicate depends on the relative position or overlap of the two boxes the score is clearly positive ("image helps"); where the class pair alone already determines the likely predicate it is near zero ("prior suffices"). The figure makes visible what the decomposition asserts: WAGER does not reward a confident prediction as such, only one that beats the model's own frequency-collapsed self.

4.4 Where reasoning concentrates, and whether gains are substantive

An aggregate I^{reason} can hide *where* a model reasons, so we stratify the per-instance scores by predicate-frequency tier (head: top 5 predicates; body: next 11; tail: remaining 34). The mean score $\bar{d}$ per tier (Figure 9, Table 4) reveals a striking and consistent pattern: head predicates are essentially won by the prior—for the spatial MLPs $\bar{d} \leq 0.026$, and for FREQ it is even slightly negative—whereas the body and tail carry almost all of the reasoning signal, with $\bar{d} = 0.16$–0.19 for MLP-SPATIAL and MLP-SPATIAL+. The small aggregate reasoning channel is therefore not spread uniformly; it is concentrated on precisely the rare predicates that headline recall metrics underweight. This is the WAGER-native restatement of a familiar concern, that aggregate scores mask tail behaviour [9, 18], now expressed as a distribution-free per-tier measurement rather than a re-weighted metric.

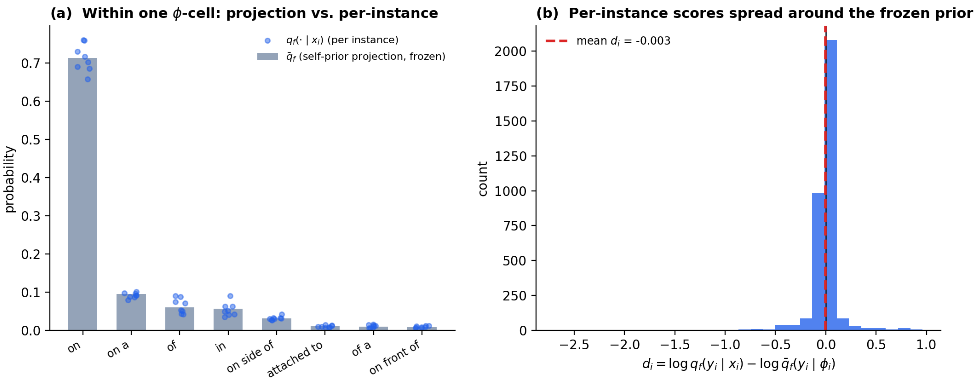

Fig. 7 Self-prior projection in one ϕ-cell. (a) The frozen projection $\bar{q}_f$ (bars) versus per-instance model predictions $q_f(\cdot \mid x_i)$ (dots). (b) The distribution of per-instance reasoning scores d_i: their spread around the prior is the signal WAGER measures.

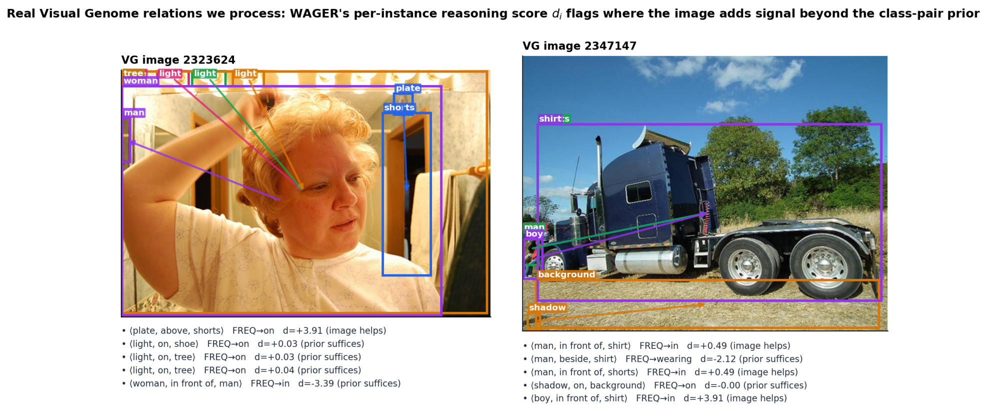

Fig. 8 Real processed examples. Visual Genome images with our overlaid subject/object boxes and relation arrows. Each caption lists the annotated triplet, the FREQ top-1 guess, and the per-instance reasoning score d_i for MLP-SPATIAL: positive d_i marks relations where the image adds signal beyond the class-pair prior.

The decisive question—whether a reported gain is substantive—is answered by the Reasoning Gain Ratio (Figure 10, Table 5). Every accuracy gain over the frequency baseline is flagged *non-substantive*, its RGR upper limit well below 0.5, so most of each reported improvement is prior-recoverable rather than reasoning. The decision rule is not vacuous: it fires positively when warranted. The step from MLP-CLASS to MLP-SPATIAL, which adds genuine box geometry, is *reasoning-supported* (RGR $= 0.77$, CI $[0.74, 0.79]$), whereas the further step to MLP-SPATIAL+ adds capacity but little new visual signal and is again non-substantive (RGR $= 0.44$). WAGER thus separates

Table 4 Mean per-instance reasoning score $\bar{d}$ (nats) by predicate-frequency tier. FREQ stays near zero in every tier; the spatial MLPs concentrate their reasoning on body and tail predicates.

Model	head	body	tail
FREQ	−0.021	+0.033	+0.038
FREQ+OVERLAP	−0.032	+0.025	+0.017
MLP-CLASS	+0.014	+0.069	+0.057
MLP-SPATIAL	+0.024	+0.162	+0.157
MLP-SPATIAL+	+0.026	+0.189	+0.182

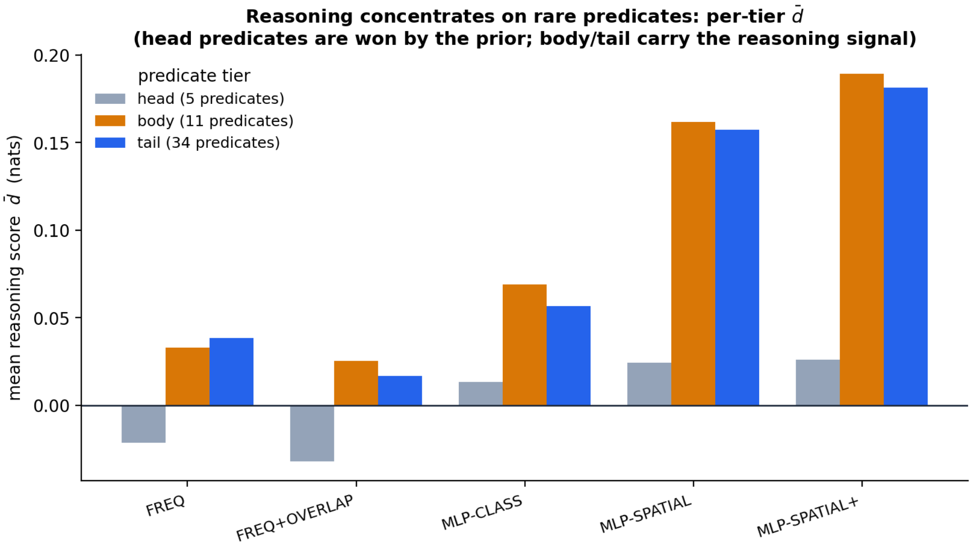

Fig. 9 **Reasoning concentrates on rare predicates.** Mean reasoning score $\bar{d}$ per predicate tier. Head predicates are won by the prior; the body and tail carry the reasoning signal, and it grows with spatial capacity.

a gain that comes from new image information from one that comes from better prior-fitting or extra capacity—a distinction that raw accuracy, varying by under 1.3 points across all five models, cannot draw.

4.5 Ablations

Finally we ask whether the headline verdict survives the principal design choices, using the MLP-SPATIAL-versus-FREQ gain as the test case (Figure 11). The verdict is stable across the prior-feature granularity (cell versus subject-only), the shrinkage pseudo-count $m \in \{0, 0.25, 1\}$, and the number of permutations $R \in \{10, 30, 60\}$: all

Table 5 Reasoning Gain Ratio with Fieller CI and verdict on VG PredCls.

Gain f' vs f	RGR (CI)	Verdict
FREQ+OVERLAP vs FREQ	$0.20\,[0.17, 0.22]$	non-substantive
MLP-CLASS vs FREQ	$0.17\,[0.16, 0.18]$	non-substantive
MLP-SPATIAL vs FREQ	$0.28\,[0.27, 0.29]$	non-substantive
MLP-SPATIAL+ vs FREQ	$0.29\,[0.28, 0.30]$	non-substantive
MLP-SPATIAL+ vs MLP-SPATIAL	$0.44\,[0.39, 0.49]$	non-substantive
MLP-SPATIAL vs MLP-CLASS	$\mathbf{0.77}\,[\mathbf{0.74}, \mathbf{0.79}]$	**reasoning-supported**

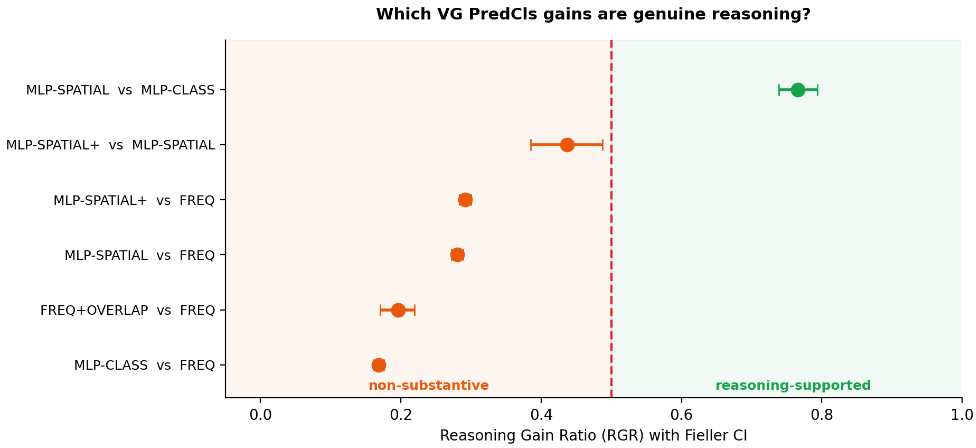

Fig. 10 **Gain attribution.** RGR with Fieller CIs; the dashed line is the 0.5 decision threshold. Only the MLP-CLASS→MLP-SPATIAL step, which adds real box geometry, is reasoning-supported.

remain non-substantive. The one sensitive knob is the clip c: under the most aggressive clipping ($c = \log K/2$) the ratio crosses 0.5, because clipping compresses the large prior channel more than the small reasoning channel, so we report the conservative default $c = \log K$ throughout. The stability under ϕ coarsening is the most important of these checks, since a missing frequency channel in ϕ would move the verdict when ϕ is coarsened, and it does not.

5 Discussion, Limitations, and Future Work

WAGER provides a specific form of evidence: it measures the usable information that a model extracts beyond a stated annotation-frequency prior and determines how much of the gain between two models belongs to that residual channel. This scope is important when interpreting either verdict. The estimated I^{reason} is family-relative in the sense of usable-information analyses [22]; it describes what the evaluated model extracts, rather than the information-theoretic limit of the task. Consequently, a non-substantive verdict applies to the selected pair of forecasters and does not imply that the task cannot be solved through reasoning. Conversely, a reasoning-supported verdict

Fig. 11 Verdict stability. RGR (MLP-SPATIAL vs FREQ) across design choices; dashed line is the 0.5 threshold. The verdict is stable except under the most aggressive clip $c = \log K/2$.

certifies predictive information beyond ϕ, but does not by itself establish correctness, causal understanding, or robustness under other distribution shifts.

This distinction positions WAGER within the broader concern that static leaderboards can overstate capability through frequency shortcuts, contamination, or memorization, particularly for multimodal systems [19–21]. Most proposed remedies refresh, randomize, or otherwise replace the benchmark. WAGER instead retains the benchmark and audits a reported gain against an explicit confound. These strategies are complementary: a dynamically refreshed or contamination-resistant benchmark may still contain annotation-frequency structure, and its model comparisons can be analyzed with WAGER to separate prior fitting from residual reasoning.

Several conditions delimit the validity of that analysis. First, the betting result requires exchangeability rather than independent and identically distributed observations. We preserve the dependence among relations from the same image by permuting complete image blocks, preventing individual relations from the same scene from being treated as independent units. Second, exact validity depends on fitting the self-prior projection on a fold independent of the betting stream. Visual Genome's sparse class-pair cells make this estimation difficult, so the projection uses Krichevsky–Trofimov smoothing, hierarchical backoff, and a Jensen correction (Table 1 and Section 3). Third, ϕ must represent the frequency channel being audited. An omitted confound may be incorrectly credited to reasoning; for that reason, we report the finest, most conservative specification of ϕ and verify that the verdict remains stable when it is coarsened (Section 4.5). Fourth, clipping controls the bounded score required by the wealth process and therefore trades robustness against sensitivity. The conclusion is stable at the default $c = \log K$ and changes only under the aggressive clipping examined in the ablation. Finally, because the measured quantity is model-relative, WAGER compares concrete frozen forecasters rather than architectures in the abstract. The cached-probability interface is what makes this comparison exact and reproducible.

The same interface also makes broader evaluation practical: a new application needs only a one-time probability dump and an appropriate definition of ϕ. In scene graph generation, applying the present pipeline to released MOTIFS, VCTree, and causal +TDE checkpoints would reveal whether published improvements survive the RGR criterion and whether TDE increases the reasoning share as its causal motivation suggests. In VQA and multimodal language models, question type can serve as ϕ, allowing gains over classical baselines to be separated into visual and language-prior contributions, where dependence on the latter remains well documented [4, 5, 17]. For LVIS-style detection, frequency tier provides the analogous prior feature, and the head/body/tail analysis of Section 4.4 extends directly to rare-category evaluation [6, 18]. Across these settings, the statistical construction remains unchanged: the self-prior projection defines the competing forecaster, one wealth process certifies and quantifies the residual signal, and RGR attributes the gain between systems.

6 Conclusion

We introduced WAGER, a distribution-free, anytime-valid framework that decomposes a computer-vision benchmark gain into a prior-recoverable part and a genuine-reasoning part with a finite-sample certificate. Its technical core is the self-prior projection—comparing a model to its own frequency-collapsed self—which makes the null exact without knowing the true prior and the measurement provably calibration-invariant. A single testing-by-betting capital process built on the projection both certifies (an e-value, via Ville's inequality) and quantifies (a growth rate equal to usable reasoning information) reasoning beyond annotation priors, and two such processes combine into the Reasoning Gain Ratio with an anytime-valid interval and an operational decision rule. On simulations WAGER controls type-I error, attains nominal coverage, and recovers injected reasoning with Pearson $r = 0.998$; on a real Visual Genome study over 7.6×10^5 relations it certifies the frequency baseline as non-reasoning and shows that every accuracy gain over it is statistically indistinguishable from improved annotation-frequency-prior fitting, while still recognizing a genuine geometry-driven gain when one is present. By casting benchmark evaluation as a game-theoretic e-process, WAGER turns "is this gain real?" into a reproducible, certified measurement. The framework operates on cached probability vectors and instantiates across SGG, VQA, and LVIS through a single design choice, the prior feature ϕ.

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Appendix A Full proofs

A.1 Theorem 1 (anytime-valid certificate)

Let $\mathcal{F}_i = \sigma(\tilde{d}_1, \ldots, \tilde{d}_i)$ and let λ_i be $\mathcal{F}_{i-1}$-measurable with $0 \leq \lambda_i < 1/c$. Because $\tilde{d}_i \in [-c, c]$ we have $1 + \lambda_i \tilde{d}_i \geq 1 - \lambda_i c > 0$, so $W_n \geq 0$. Under H_0, $\mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}] \leq 0$, hence

$$\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] = W_{i-1}\big(1 + \lambda_i \, \mathbb{E}[\tilde{d}_i \mid \mathcal{F}_{i-1}]\big) \leq W_{i-1},$$

so (W_n) is a non-negative supermartingale with $\mathbb{E}[W_n] \le W_0 = 1$. Ville's inequality [41] for non-negative supermartingales states $\Pr[\sup_n W_n \ge 1/\alpha] \le \mathbb{E}[W_0]\,\alpha = \alpha$, which is the claimed time-uniform type-I bound; stopping the first time $W_n \ge 1/\alpha$ therefore rejects at level α regardless of the (possibly data-dependent) stopping rule. $\square$

A.2 Theorem 2 (growth rate equals usable reasoning information)

Write $g_i = \log(1 + \lambda_i \tilde{d}_i)$ so $\frac{1}{n} \log W_n = \frac{1}{n} \sum_i g_i$. For the log-optimal predictable bet $\lambda_i^\star = \arg\max_\lambda \mathbb{E}[\log(1 + \lambda \tilde{d}) \mid \mathcal{F}_{i-1}]$, the Kelly/Cover–Thomas theory of horse-race and forecasting games [37–39] gives that the optimal expected log-growth equals the expected log-score differential of the two forecasters. Under exchangeability the strong law yields $\frac{1}{n} \sum_i \tilde{d}_i \to \mathbb{E}[\tilde{d}]$ a.s., and the optimized growth rate converges to the same limit. Taking $c \to \infty$ replaces $\tilde{d}$ by $d = \log q_f(Y \mid X) - \log \bar{q}_f(Y \mid \phi(X))$, so $G(f) = \mathbb{E}[\log q_f(Y \mid X)] - \mathbb{E}[\log \bar{q}_f(Y \mid \phi(X))] =: \mathcal{I}_f^{\text{reason}}$. Non-negativity follows because $\bar{q}_f$ is the ϕ-measurable projection of q_f and Gibbs' inequality makes the conditional log-loss of the coarser forecaster at least that of the finer one. If $q_f = p(\cdot \mid x)$ and $\bar{q}_f = p(\cdot \mid \phi)$, then $\mathcal{I}_f^{\text{reason}} = \mathbb{E} \log \frac{p(Y|X)}{p(Y|\phi(X))} = I(Y; X \mid \phi(X))$ by the chain rule $I(Y;X) = I(Y;\phi(X)) + I(Y;X \mid \phi(X))$, valid since ϕ is a function of X. $\square$

A.3 Corollary 1 (calibration invariance)

Let T_ϕ be any map applied uniformly to all predictions in cell ϕ (e.g. per-cell temperature scaling), and let $q_f^T(\cdot \mid x) = T_{\phi(x)}(q_f(\cdot \mid x))$. Its projection is $\bar{q}_f^T(\cdot \mid \phi) = T_\phi(\bar{q}_f(\cdot \mid \phi))$ because the projection averages within the cell where T_ϕ is constant in form. In the difference $d^T = \log q_f^T(y \mid x) - \log \bar{q}_f^T(y \mid \phi)$ the common transformation cancels up to the within-cell-constant Jacobian term, leaving $\mathbb{E}[d^T] = \mathbb{E}[d]$, so $\mathcal{I}_f^{\text{reason}}$ is invariant. Moreover, for an *arbitrary* miscalibrated q_f, Gibbs' inequality gives $\mathbb{E}[\log q_f(Y \mid X)] \le \mathbb{E}[\log p(Y \mid X)]$ with the analogous bound for $\bar{q}_f$, so miscalibration can only shrink the measured reasoning, never inflate it above the true value. $\square$

A.4 Theorem 3 (anytime-valid CI for RGR)

By [31], the hedged capital process gives an anytime-valid $(1-\alpha)$ confidence sequence for the mean of a bounded variable; applied to the clipped numerator score $\Delta d = d_{f'} - d_f$ and denominator score $\Delta(d+e) = (d_{f'} + e_{f'}) - (d_f + e_f)$ it yields intervals $\mathcal{N}_{1-\alpha}$ and $\mathcal{D}_{1-\alpha}$ valid simultaneously over n. By a union bound the event $\{\mathbb{E}[\Delta d] \in \mathcal{N}\} \cap \{\mathbb{E}[\Delta(d+e)] \in \mathcal{D}\}$ has probability $\ge 1 - 2\alpha$. On that event Fieller's construction [42] maps the numerator/denominator interval pair to a set containing $\text{RGR} = \mathbb{E}[\Delta d]/\mathbb{E}[\Delta(d+e)]$; the set is a bounded interval when the denominator interval excludes 0 and is the whole line otherwise. The result is a $(1-2\alpha)$ anytime-valid interval for RGR. $\square$

Appendix B Implementation details

Online-Newton-Step (aGRAPA) bet.

With most recent observation z, previous fraction λ, and accumulated curvature A (initialized $A_0 = 1$), the predictable update of [31] is

$$g = \frac{z}{1 + \lambda z}, \quad A \leftarrow A + g^2, \quad \lambda \leftarrow \mathrm{clip}\Big(\lambda + \tfrac{2}{2 - \ln 3}\, \tfrac{g}{A},\, 0,\, \tfrac{1}{2c}\Big).$$

The cap at $1/(2c)$ (rather than $1/c$) gives numerical headroom while preserving $1 + \lambda \tilde{d} > 0$.

Hierarchical projection and Jensen correction.

For each cell the shrunk mean toward a parent distribution π is $\hat{\bar{q}} = \frac{\sum_j q_j + m\pi}{n+m}$, with hierarchy cell→coarse (subject)→global. Estimating $\log \bar{q}$ by $\log \hat{\bar{q}}$ is biased downward by Jensen's inequality; we add the second-order delta-method correction $\frac{1}{2}\,\widehat{\mathrm{Var}}[\hat{\bar{q}}]/\hat{\bar{q}}^2$ (capped at $\log K$), where $\widehat{\mathrm{Var}}$ is the within-cell sample variance divided by the cell count. The correction cancels in $d{+}e = I^{\mathrm{tot}}$ and only reallocates mis-credited reasoning into the prior channel.

Anytime-valid CI by hedged capital.

For a candidate mean m we run two predictable capital processes on the centered residuals $x_i - m$, one betting the residual is positive and one negative, and combine them as $W = (W^+ + W^-)/2$, itself a non-negative martingale with $\mathbb{E}[W] \leq 1$ under $H_0 : \mathbb{E}[x] = m$. The interval is $\{m : \sup_n W(m) < 1/\alpha\}$. With a predictable fixed λ estimated from a held-out prefix, the whole grid is evaluated by a vectorized cumulative sum, chunked to bound memory.

Hyperparameters.

Table B1 lists the defaults used in all experiments unless ablated.

Table B1 Default hyperparameters.

Symbol	Meaning	Value
c	clip / log-ratio scale	$\log K$
α	error level	0.05
split	projection / eval fold	40% / 60% of test images
m	shrinkage pseudo-count	1.0
R	image-blocked permutations	24
backoff	projection hierarchy	cell → subject → global

Appendix C Reproducibility

Data preparation parses the Visual Genome relationships export into a single array of (subject, object, predicate, boxes, image id, split) tuples under the VG150 vocabulary. Models are trained once on the training split and their predicate distributions on the test split are cached; WAGER then runs on the cache. Object/spatial MLPs are small (two hidden layers, dropout 0.1, AdamW) and train in minutes on a single GPU; all WAGER computation is deterministic CPU NumPy under fixed seeds. The simulation suite, the real study (Tables 3, 5), the stratified analysis (Table 4), and all figures are produced by scripts released with the code, each reading the same cached artifacts so the numbers in the paper and the figures derive from one run.